Biodiversity and Ecosystem Functioning

Today's Agenda:

- Quiz
- Ecosystem Functions and Services
- Biodiversity and Ecosystem Functioning
- BEF examples

Ecosystem Functioning

- The biological and chemical processes that occur in ecosystems
- Functions that benefit people are called “ecosystem services”
    - Also referred to as “nature’s contributions to people”

Ecosystem Services

What examples can folks think of?

Ecosystem Services

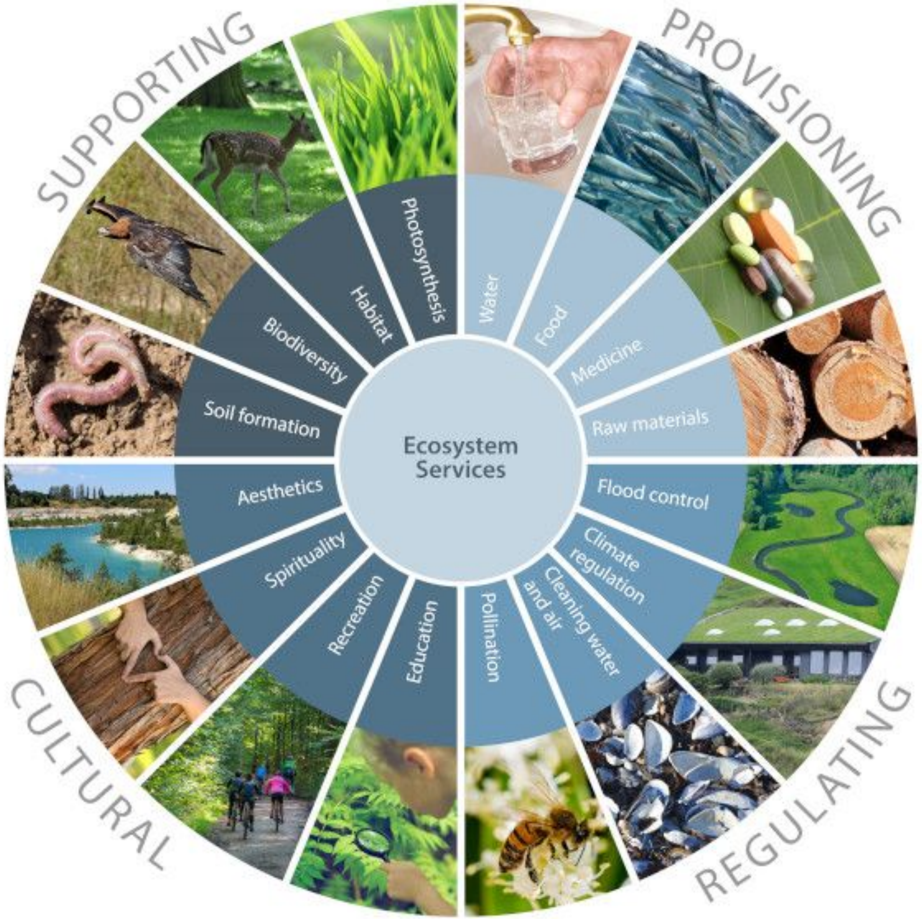


Figure: https://envs.au.dk/en/research-areas/society-environment-and-resources/biodiversity-and-ecosystem-services

Ecosystem Services

- Supporting
  - maintain fundamental ecosystem processes

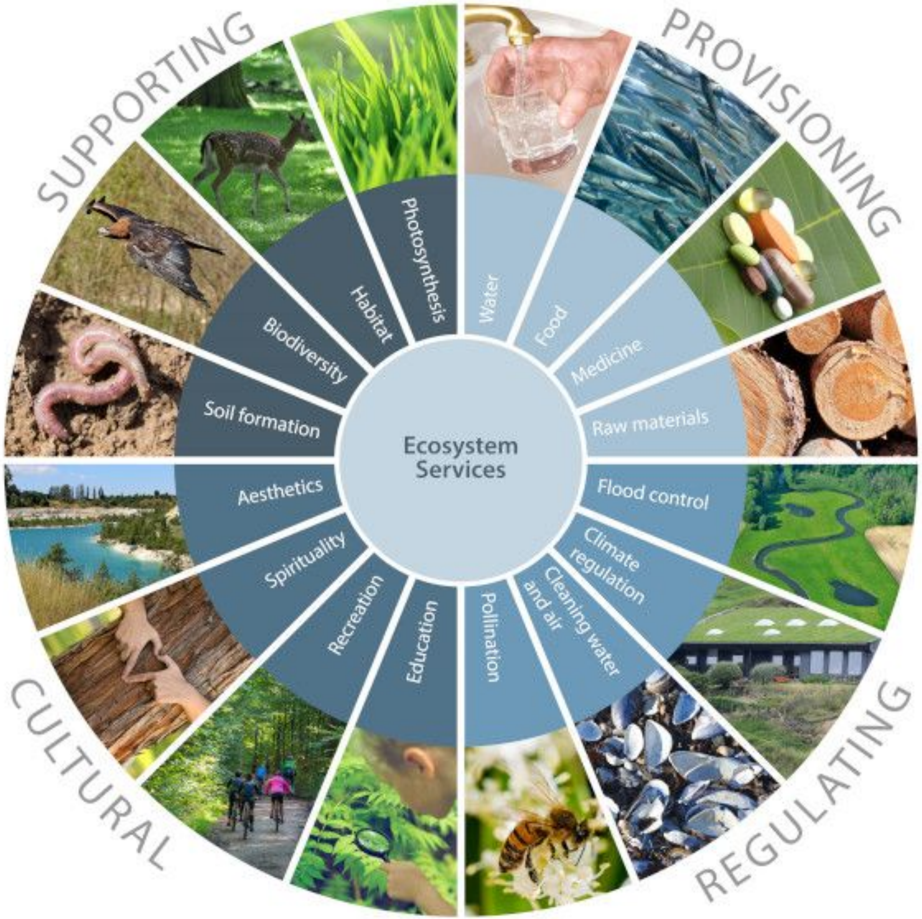
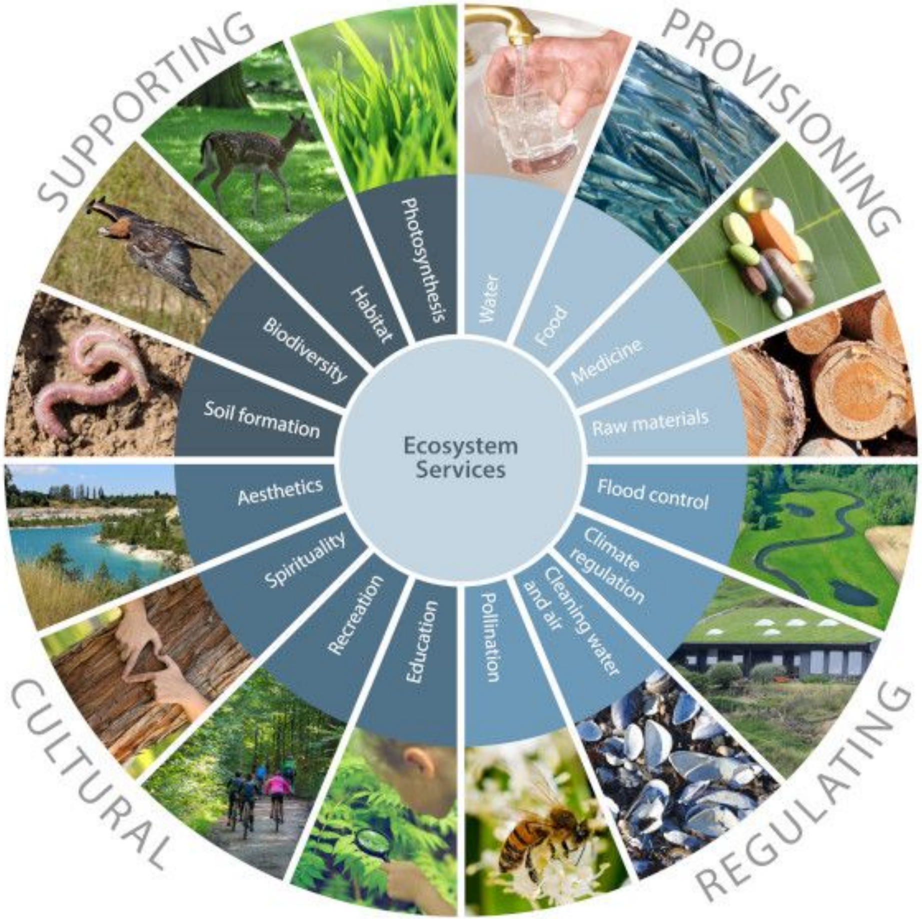


Figure: https://envs.au.dk/en/research-areas/society-environment-and-resources/biodiversity-and-ecosystem-services; definitions: https://www.climatehubs.usda.gov/ecosystem-services

Ecosystem Services

- Supporting
  - maintain fundamental ecosystem processes
- Provisioning
  - Material and energy



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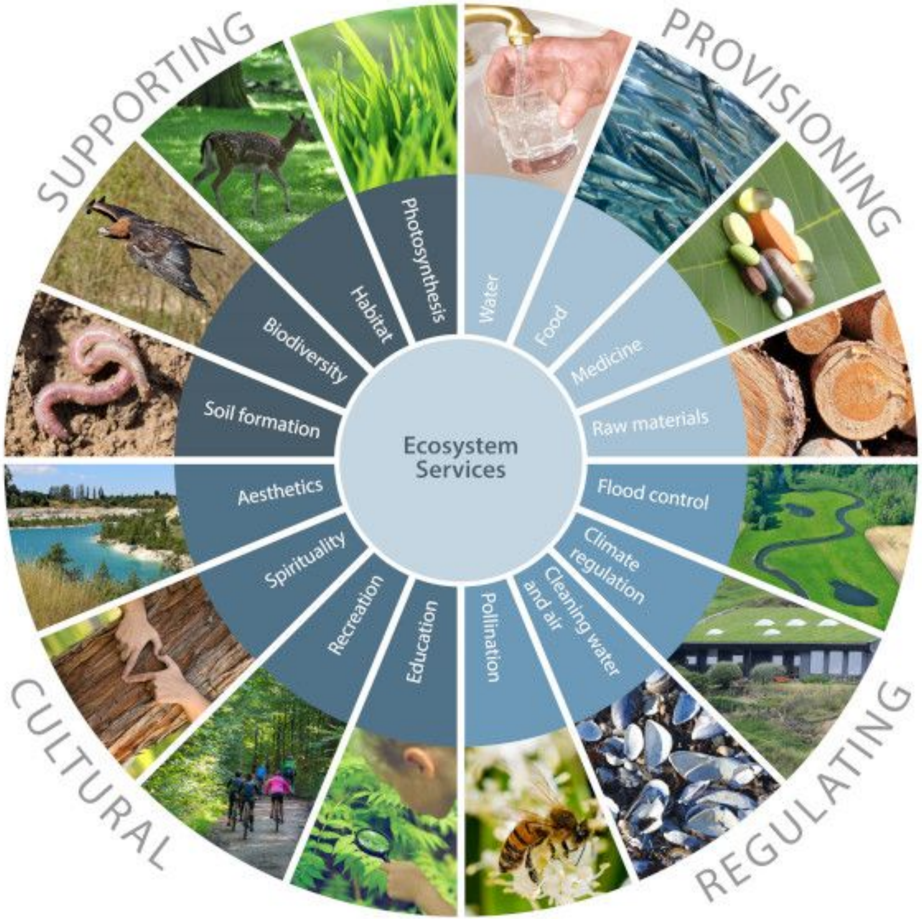
Ecosystem Services

- Supporting
  - maintain fundamental ecosystem processes
- Provisioning
  - Material and energy
- Regulating
  - moderation or control of ecosystem processes



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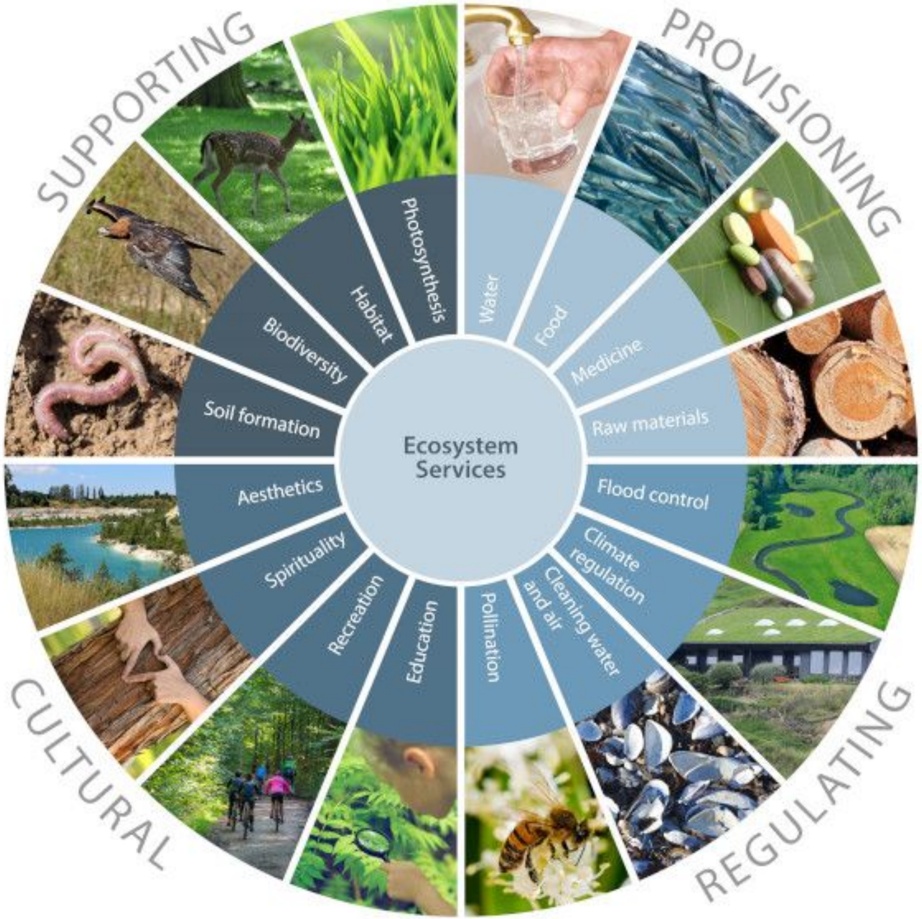
Ecosystem Services

- Supporting
  - maintain fundamental ecosystem processes
- Provisioning
  - Material and energy
- Regulating
  - moderation or control of ecosystem processes
- Cultural
  - non-material/energy benefits to human societies



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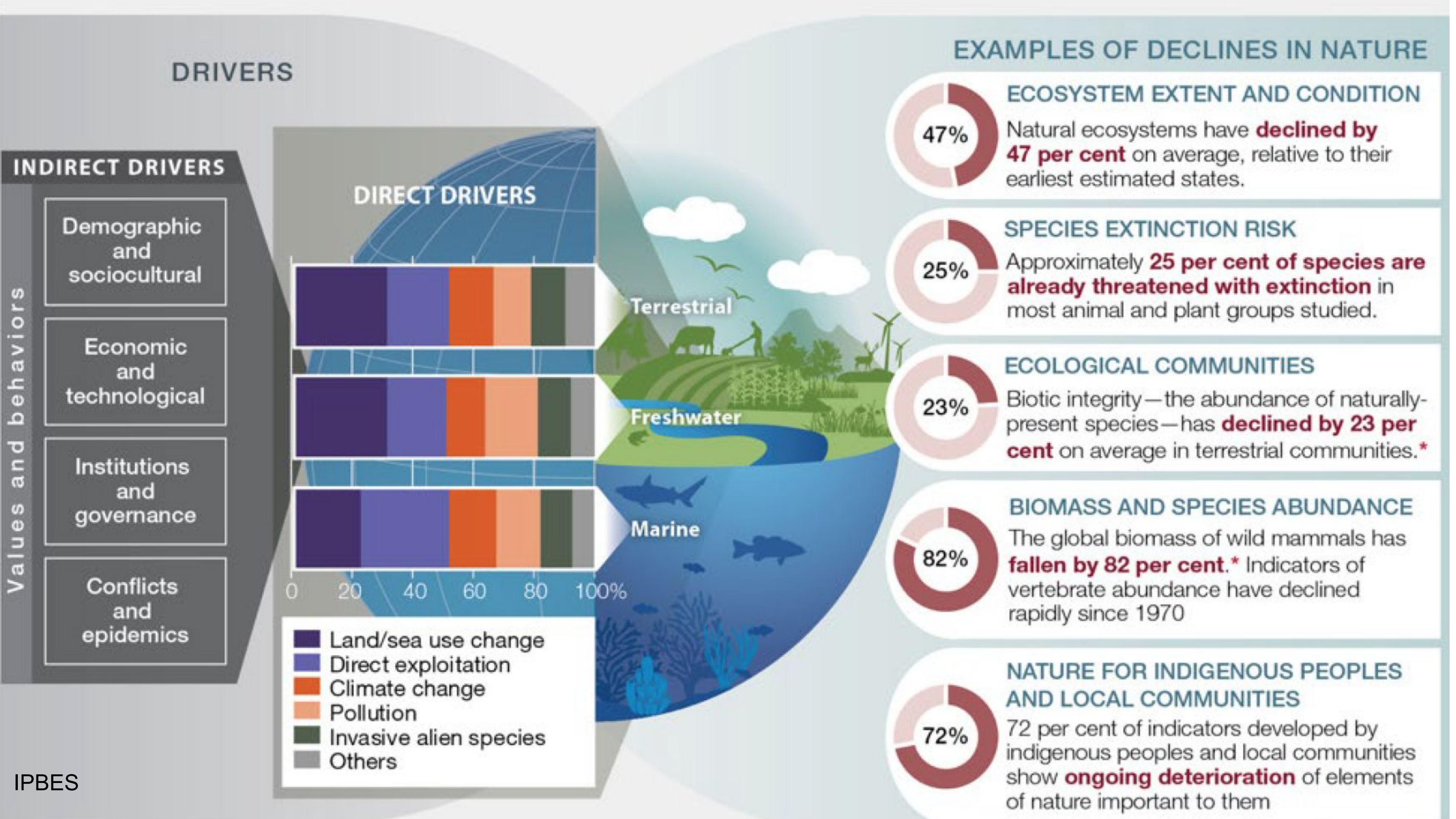
DRIVERS
INDIRECT DRIVERS
Values and behaviors
Demographic and sociocultural
Economic and technological
Institutions and governance
Conflicts and epidemics
DIRECT DRIVERS
Terrestrial
Freshwater
Marine
0 20 40 60 80 100%
Land/sea use change
Direct exploitation
Climate change
Pollution
Invasive alien species
Others
EXAMPLES OF DECLINES IN NATURE
ECOSYSTEM EXTENT AND CONDITION
47%
Natural ecosystems have declined by 47 per cent on average, relative to their earliest estimated states.
SPECIES EXTINCTION RISK
25%
Approximately 25 per cent of species are already threatened with extinction in most animal and plant groups studied.
ECOLOGICAL COMMUNITIES
23%
Biotic integrity—the abundance of naturally-present species—has declined by 23 per cent on average in terrestrial communities.*
BIOMASS AND SPECIES ABUNDANCE
82%
The global biomass of wild mammals has fallen by 82 per cent.* Indicators of vertebrate abundance have declined rapidly since 1970
NATURE FOR INDIGENOUS PEOPLES AND LOCAL COMMUNITIES
72%
72 per cent of indicators developed by indigenous peoples and local communities show ongoing deterioration of elements of nature important to them

IPBES

How much biodiversity do we need?

- Biodiversity provides things we need
- We’re destroying nature at mass extinction rates
- Need to predict impacts of our actions

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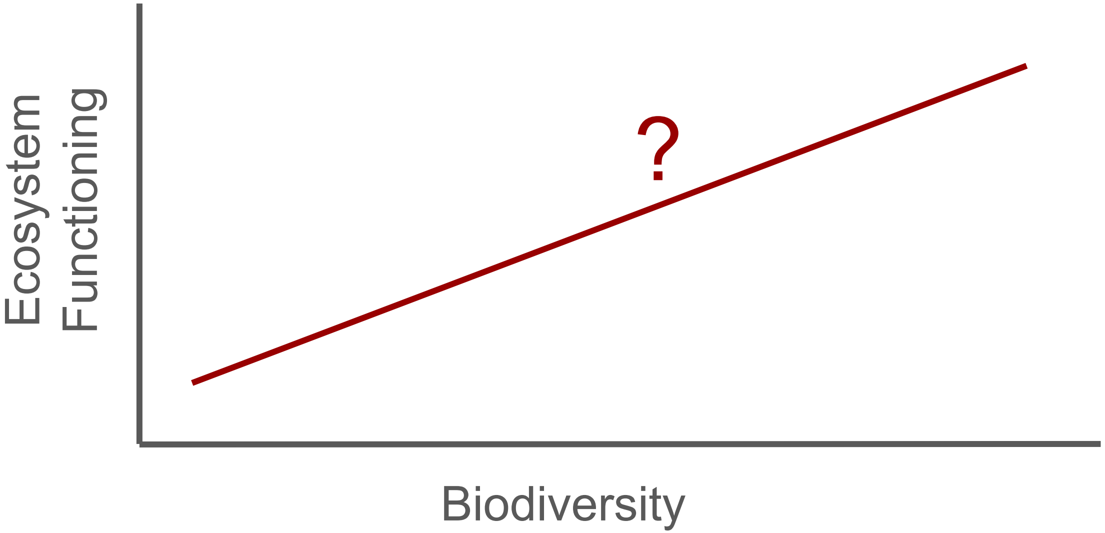

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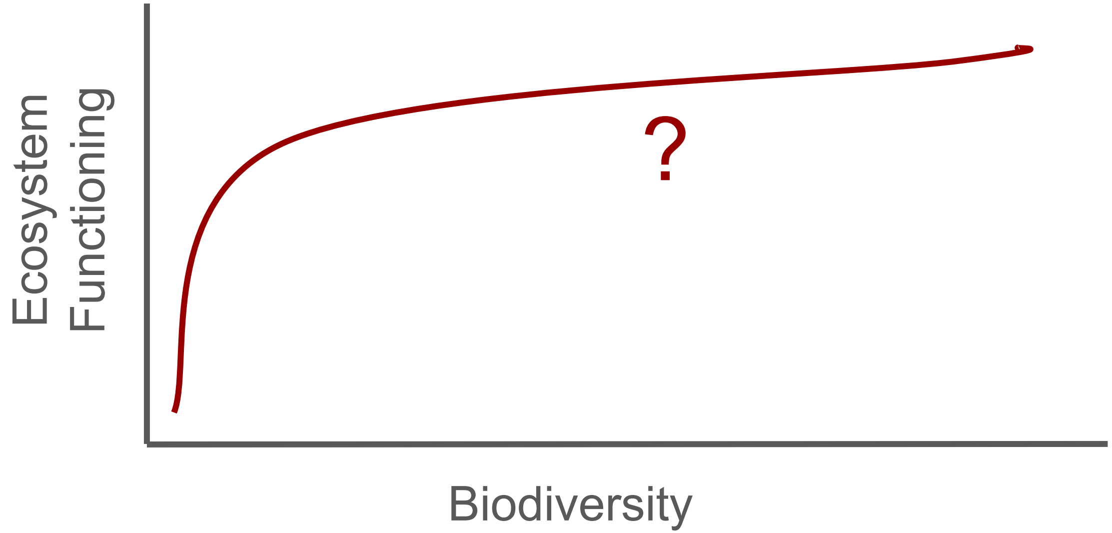

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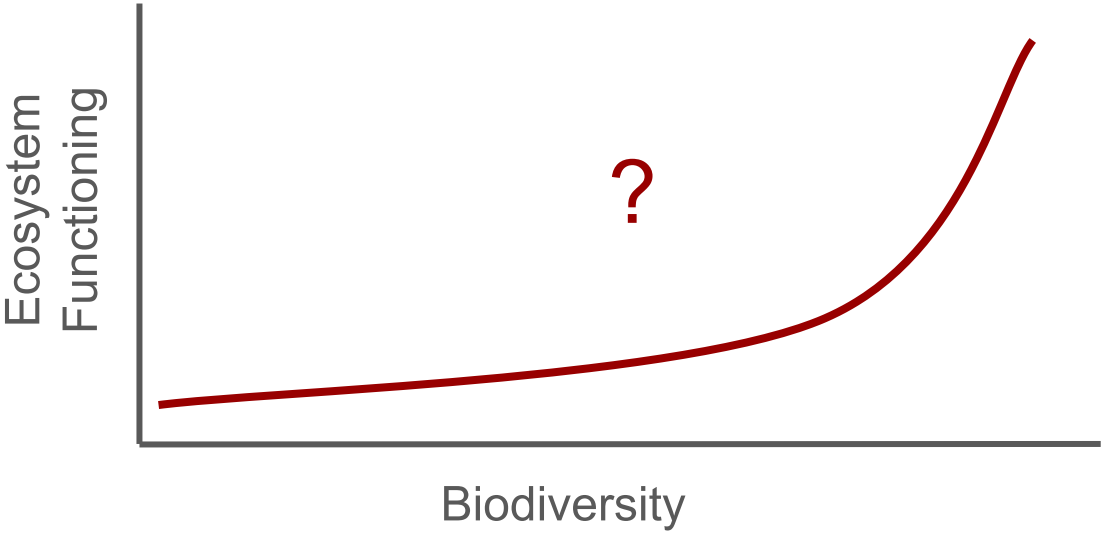
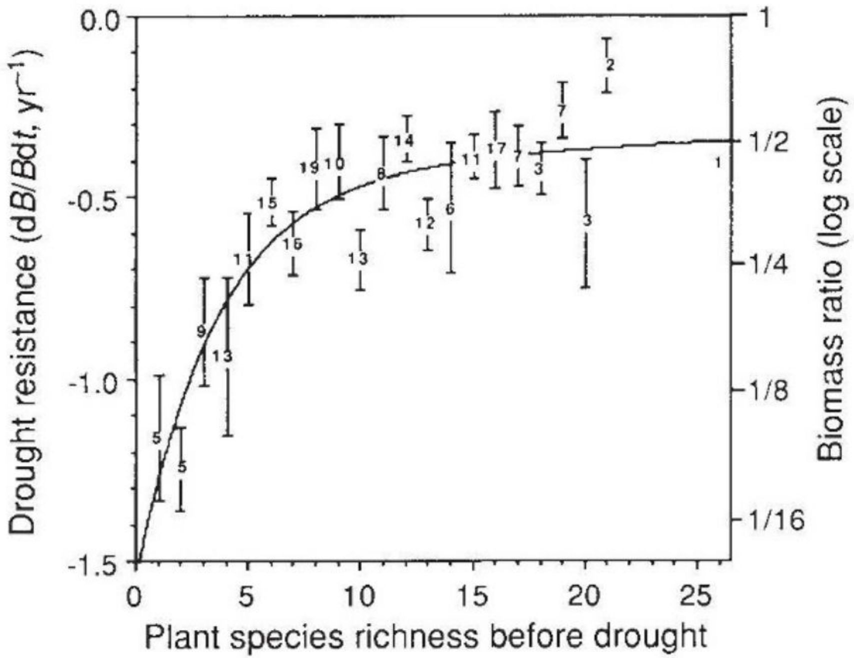


Early Experimental Evidence

- Grassland system
- Suggestive, but diversity was altered by fertilizer, confounding things

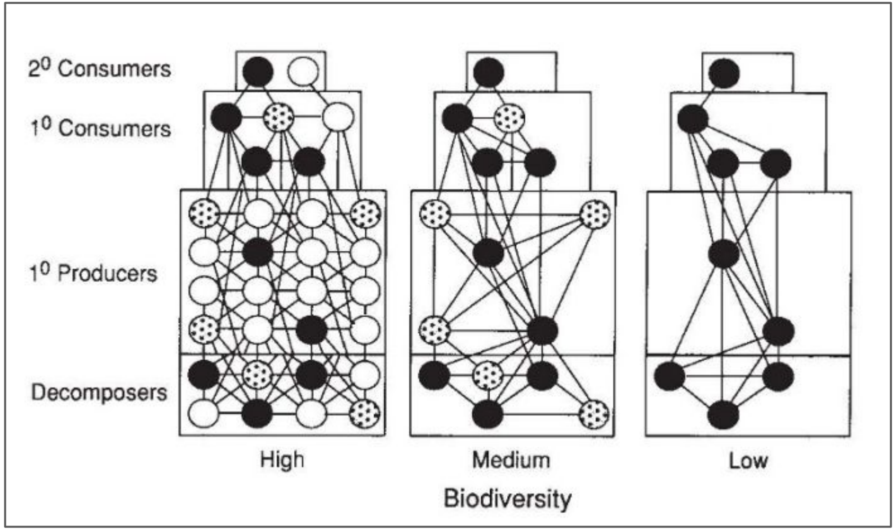


Biodiversity and stability in grasslands

David Tilman & John A. Downing

Early Experimental Evidence

- Fancy environmental chambers
- Manipulated species directly
- Manipulated multiple trophic levels



Declining biodiversity can alter the performance of ecosystems

Shahid Naeem, Lindsey J. Thompson, Sharon P. Lawler, John H. Lawton & Richard M. Woodfin

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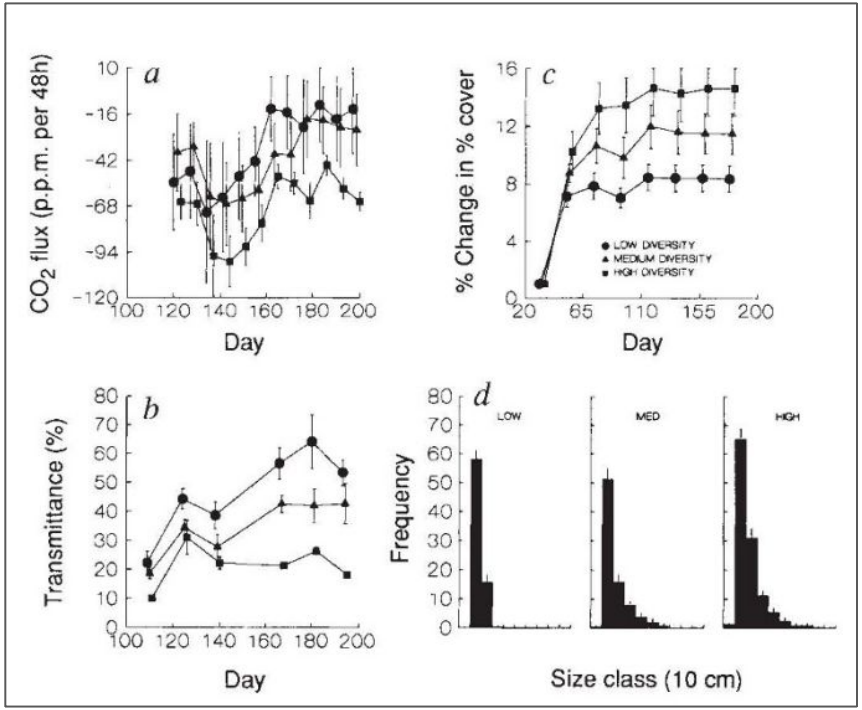

Declining biodiversity can alter the performance of ecosystems

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Focal ecosystem functions

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Diversity and Productivity

- Investigated by 3, high-profile, long-term, experiments
  - Cedar Creek (US)
  - Jena (Germany)
  - BIODEPTH (Europe)
- In all, species diversity was experimentally manipulated

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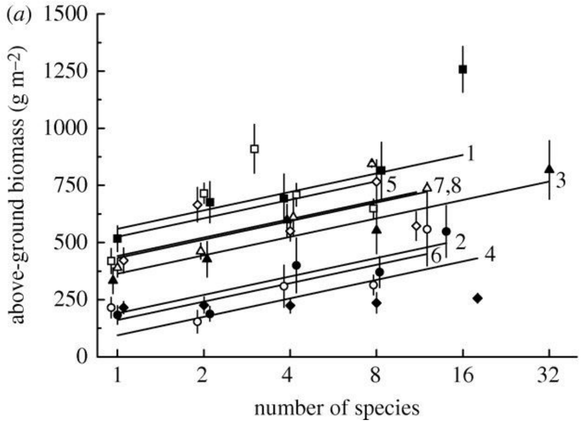

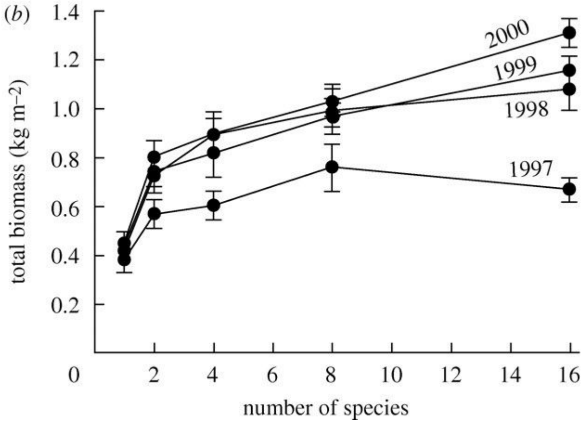


PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY B

Phil. Trans. R. Soc. B (2010) 365, 49–60
doi:10.1098/rstb.2009.0155

Review

Linking biodiversity and ecosystems: towards a unifying ecological theory

Michel Loreau*

a: BIODEPTH,
b: Cedar Creek

Diversity and Productivity

- Strong evidence that diversity drives productivity…but why?

Why Does Diversity Impact Productivity?

Two main hypotheses:

- Niche complementarity
  - Species differ in resources, predators, impacts on environment, etc.

- Sampling Effect (aka species selection)
  - More species, more likely to include highly-productive species

Testing complementarity vs sampling effects

How can we tease apart these hypotheses that make similar predictions about diversity-productivity relationships?

Testing complementarity vs sampling effects

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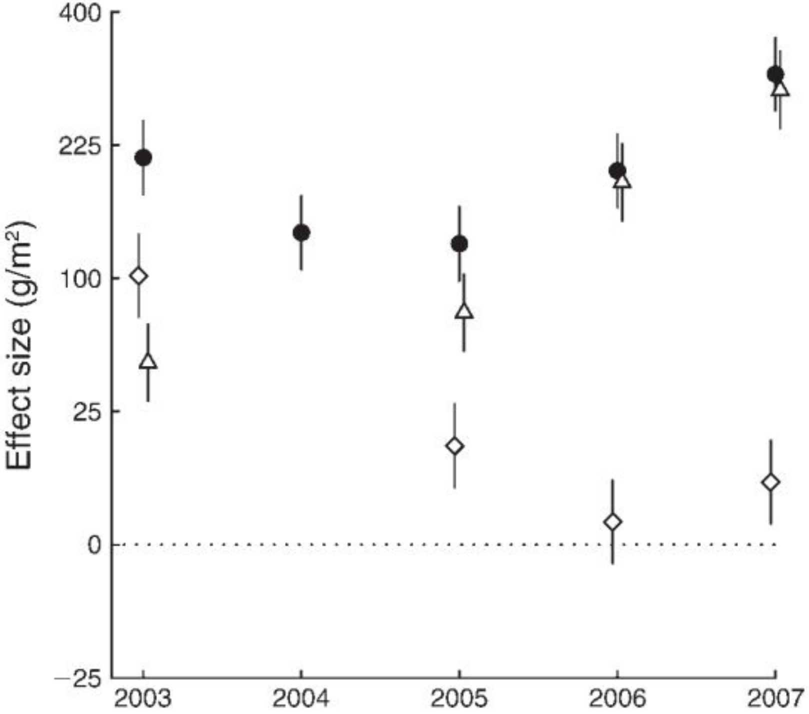
Monoculture vs polyculture experiments!

Testing complementarity vs sampling effects

- Measure services provided by a species alone
- Measure services provided by different assemblages
- Sampling effect: polyculture results should equal monoculture results weighted by abundance
- Complementarity effect: polyculture results should be more than monoculture effects
- Can partition the relative composition of these

Complementarity vs sampling



Ecology, 90(12), 2009, pp. 3290–3302
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Plant species richness and functional composition drive overyielding in a six-year grassland experiment

Elisabeth Marquard,[1,2,6] Alexandra Weigelt,[3] Vicky M. Temperton,[1,7] Christiane Roscher,[1] Jens Schumacher,[1,8] Nina Buchmann,[4] Markus Fischer,[5] Wolfgang W. Weisser,[3] and Bernhard Schmid[2]

Circles = Net Effect of Biodiversity
Diamonds = Selection
Triangles = Complementarity

Complementarity vs sampling

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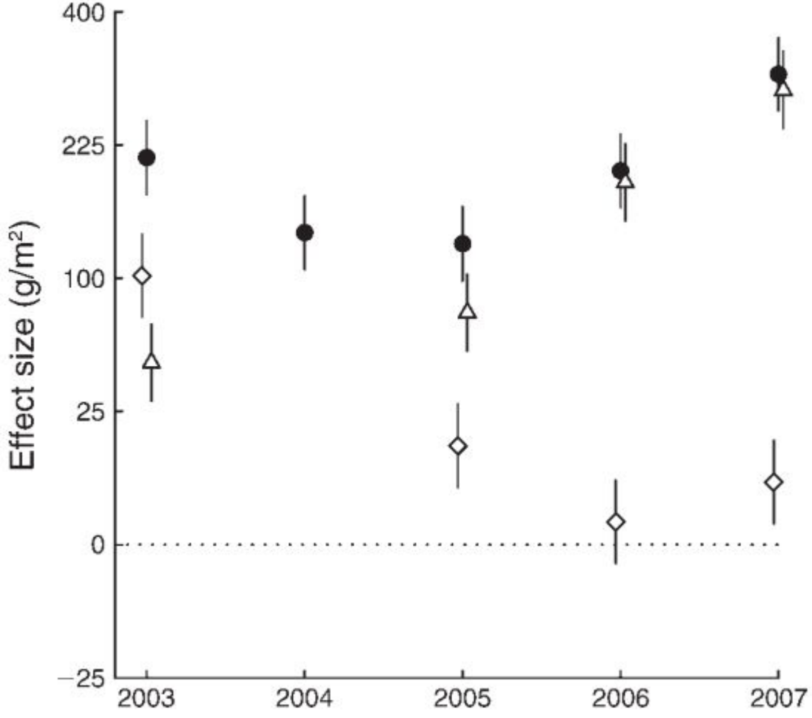

Complementarity effects increase,
Selection effects decrease

Circles = Net Effect of Biodiversity
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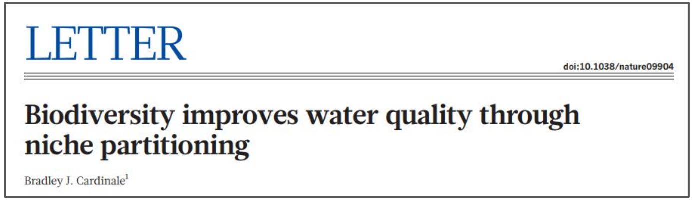
Diversity and Nutrients

- If species have complementary niches, they should use more of the available resources
- More diverse communities are more likely to contain species that use resources more efficiently
- So again, sampling and complementarity effects

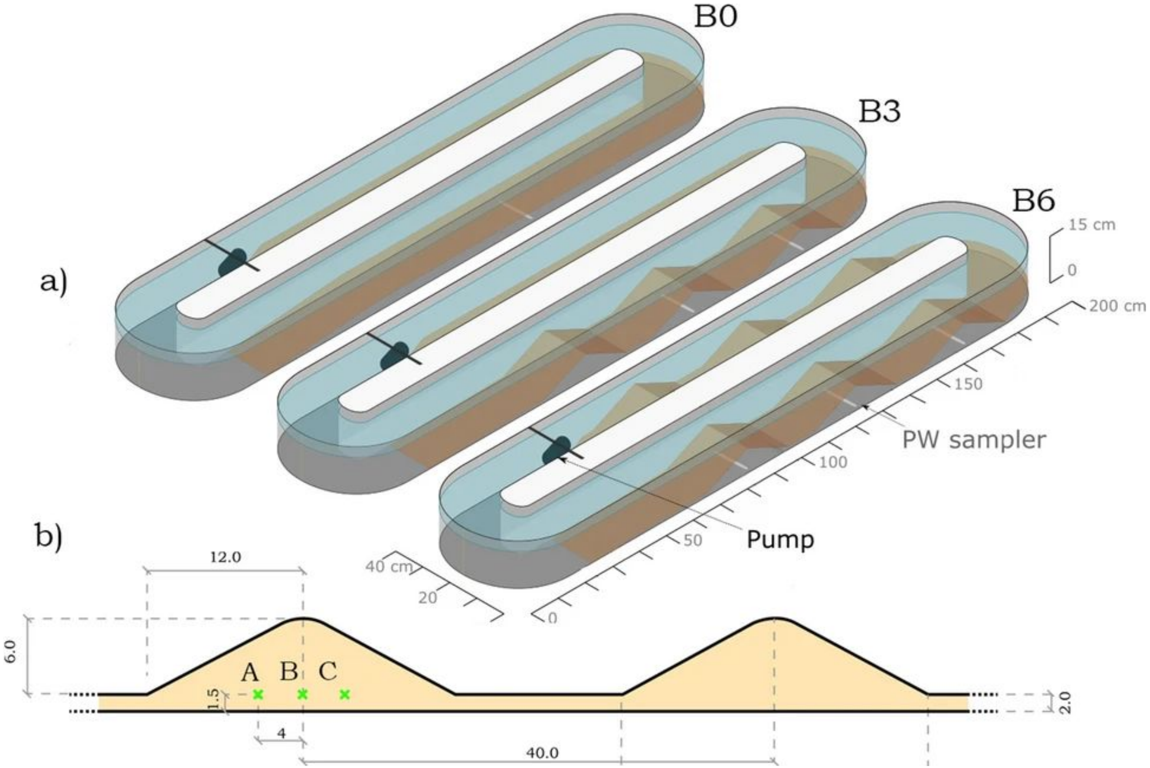
Diversity and Nutrients


LETTER

doi:10.1038/nature09904

Biodiversity improves water quality through niche partitioning

Bradley J. Cardinale[1]



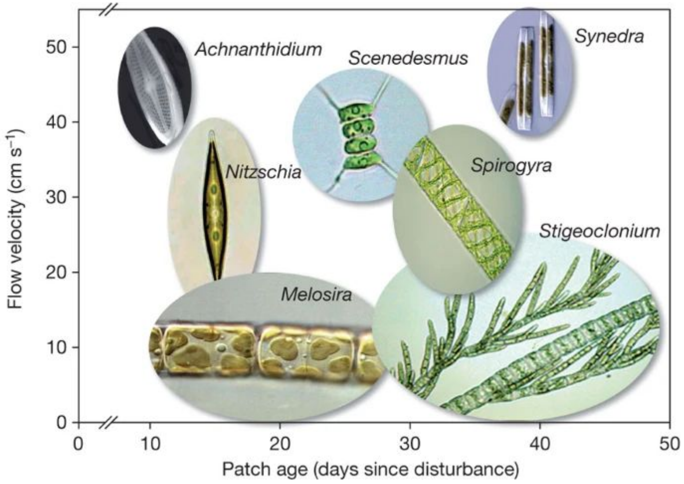
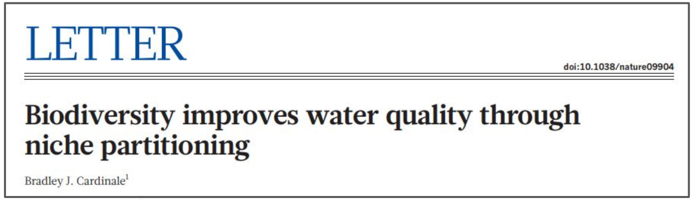


Image: Betterle, A., Jaeger, A., Posselt, M. *et al.* Hyporheic exchange in recirculating flumes under heterogeneous bacterial and morphological conditions. *Environ Earth Sci* 80, 234 (2021). https://doi.org/10.1007/s12665-021-09472-2

Diversity and Nutrients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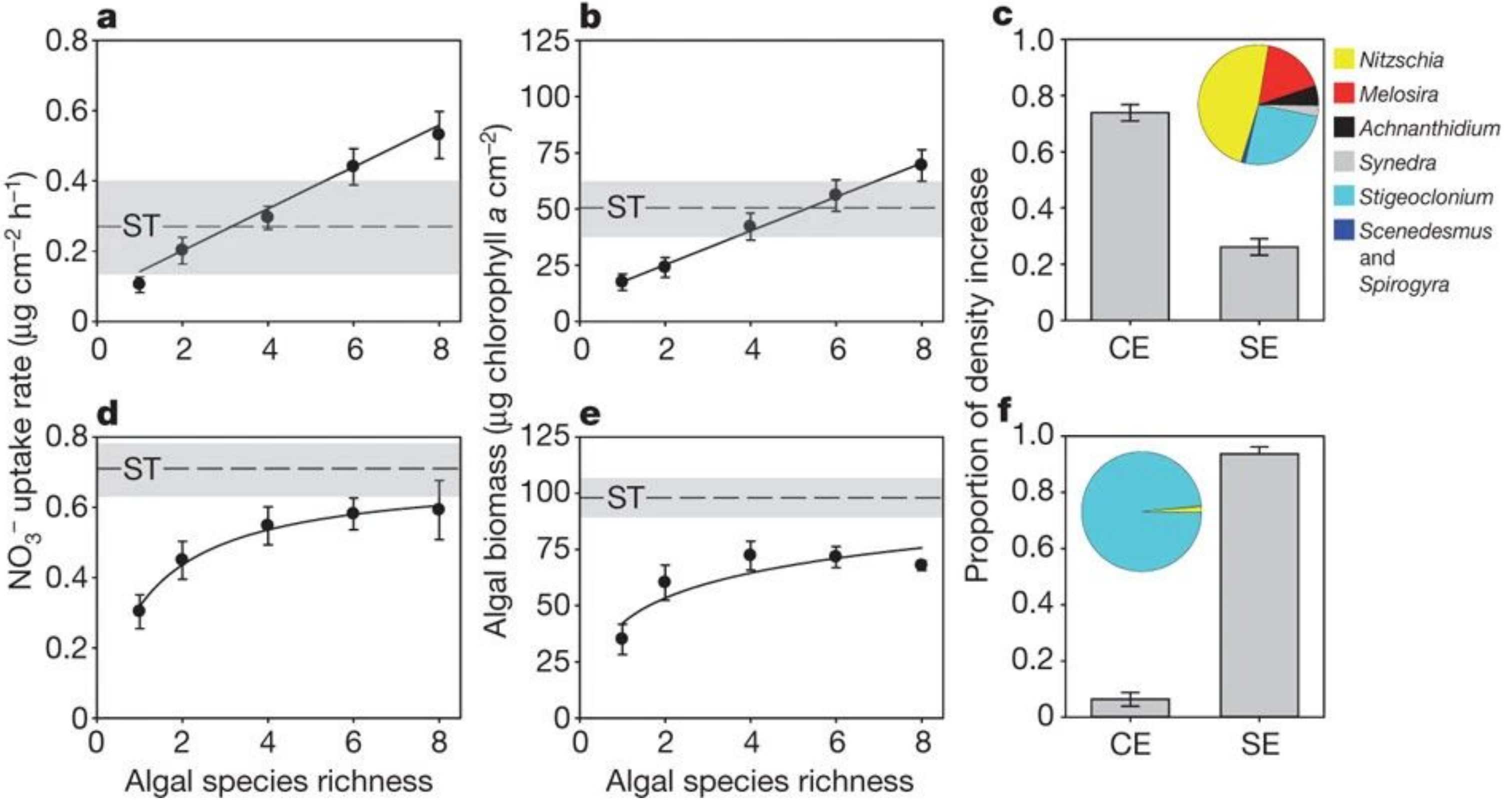
Heterogeneous streams

Homogeneous streams



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Diversity and Stability

How do we quantify stability?

- Variation in a quantity of interest over time
- Usually measured as the inverse of the coefficient of variation (CV)
  - CV = Standard Deviation/Mean
- Stability = 1/CV = Mean/Standard Deviation

Diversity and Stability

$$\text{Stability} = \frac{\text{Mean Abundance}}{\text{SD Abundance}}$$

Diversity and Stability

More individuals,
more stability

$$\text{Stability} = \frac{\text{Mean Abundance}}{\text{SD Abundance}}$$

Diversity and Stability

Less variation in abundance, more stability

$$\text{Stability} = \frac{\text{Mean Abundance}}{\text{SD Abundance}}$$

Diversity and Stability

Less variation in abundance, more stability

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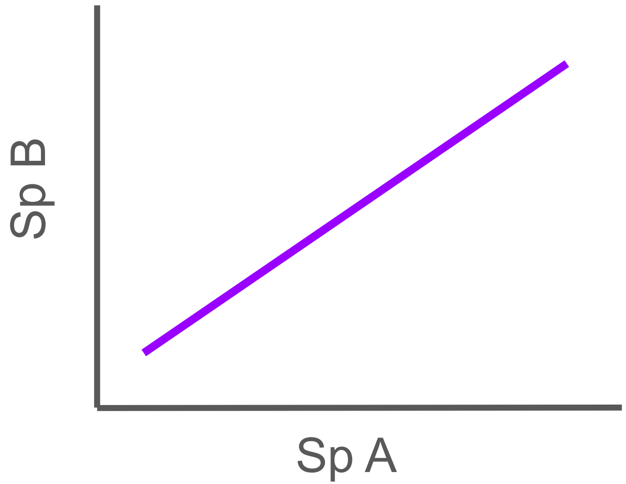
Positive covariance INCREASES SD



Diversity and Stability

Less variation in abundance, more stability

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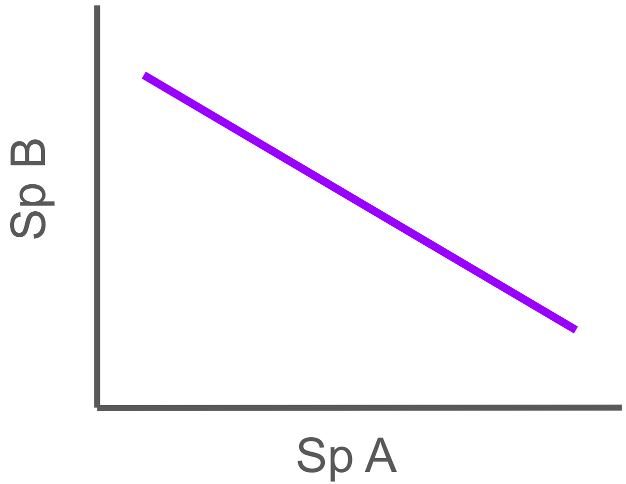


Negative covariance DECREASES SD

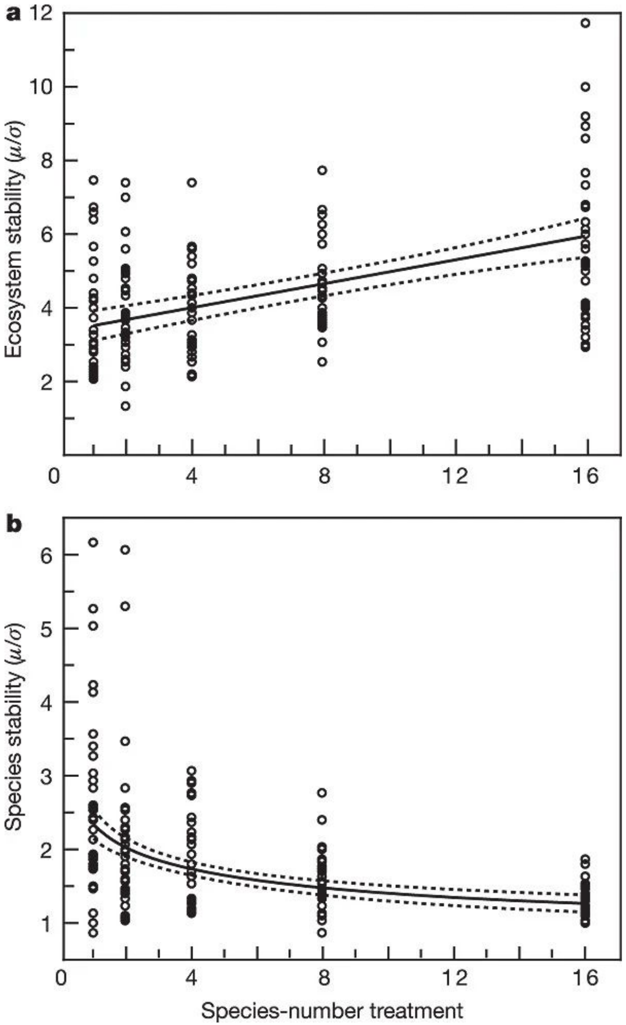
Diversity and Stability

Vol 441|1 June 2006|**doi:10.1038/nature04742** nature

LETTERS

Biodiversity and ecosystem stability in a decade-long grassland experiment

David Tilman[1], Peter B. Reich[2] & Johannes M. H. Knops[3]



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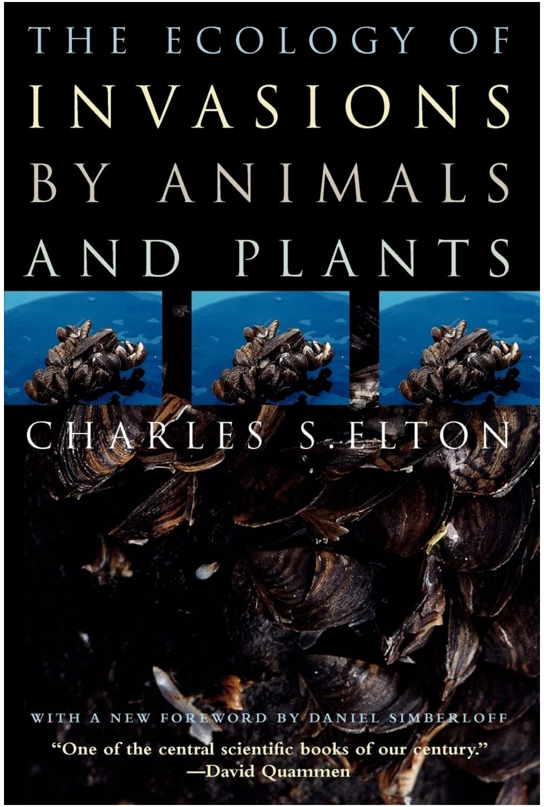
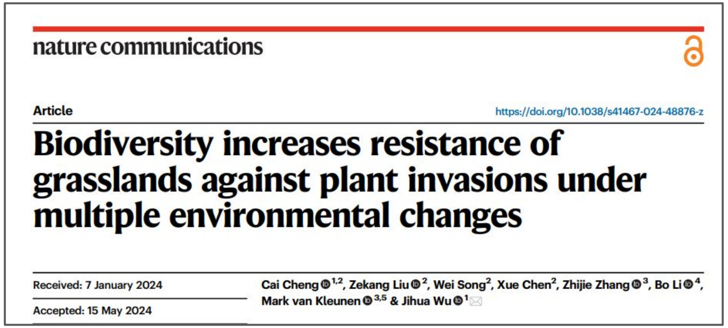
Diversity and Invasibility

Diversity and Invasibility

- More diverse = more competitors, niche overlap
- Selection effects possible - more diverse communities more likely to contain species that resist invasion



Diversity and Invasibility


nature communications

Article

https://doi.org/10.1038/s41467-024-48876-z

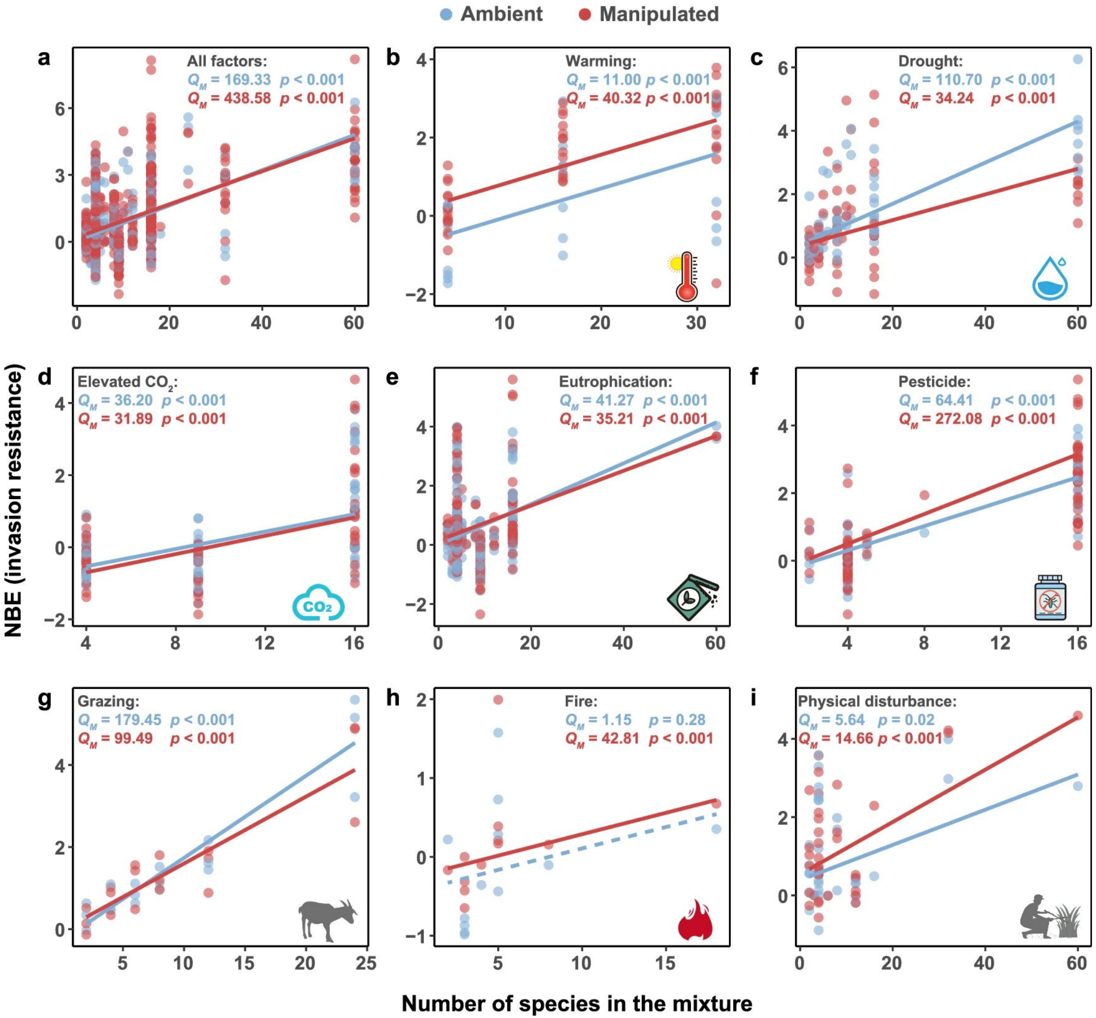
Biodiversity increases resistance of grasslands against plant invasions under multiple environmental changes

Received: 7 January 2024

Accepted: 15 May 2024

Cai Cheng[1,2], Zekang Liu[2], Wei Song[2], Xue Chen[2], Zhijie Zhang[3], Bo Li[4], Mark van Kleunen[3,5] & Jihua Wu[1]

Meta-analysis of 25 studies



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… but ecosystems provide MANY services at once…

Ecosystem Multifunctionality

- Provision of multiple functions/services by ecosystems
- Different ways of calculating
    - Standardized averages
    - Number of functions that exceed some threshold value (e.g., 50% of max functioning)

Ecosystem Multifunctionality

nature ecology & evolution

Article

https://doi.org/10.1038/s41559-024-02517-2

Plant diversity enhances ecosystem multifunctionality via multitrophic diversity

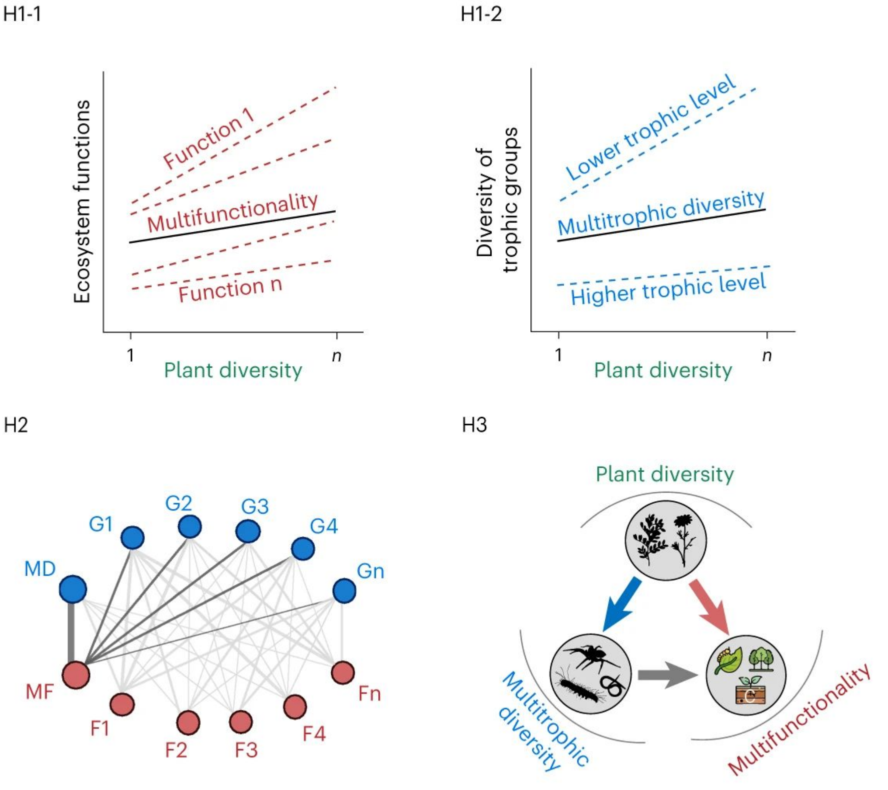
Received: 25 March 2024

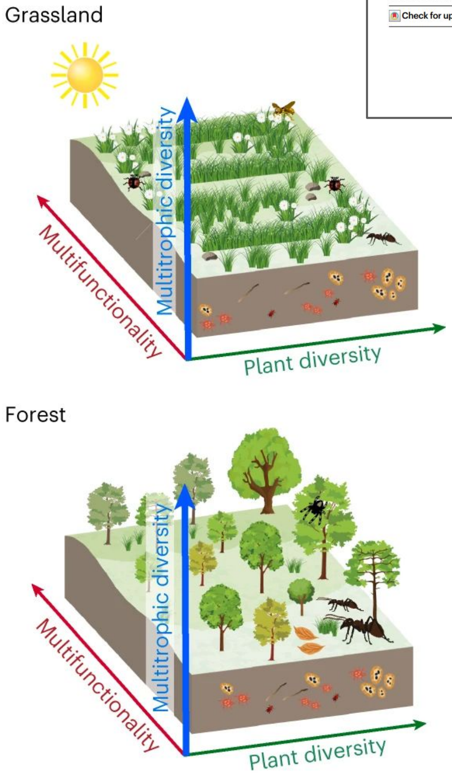
Accepted: 24 July 2024

Published online: 29 August 2024

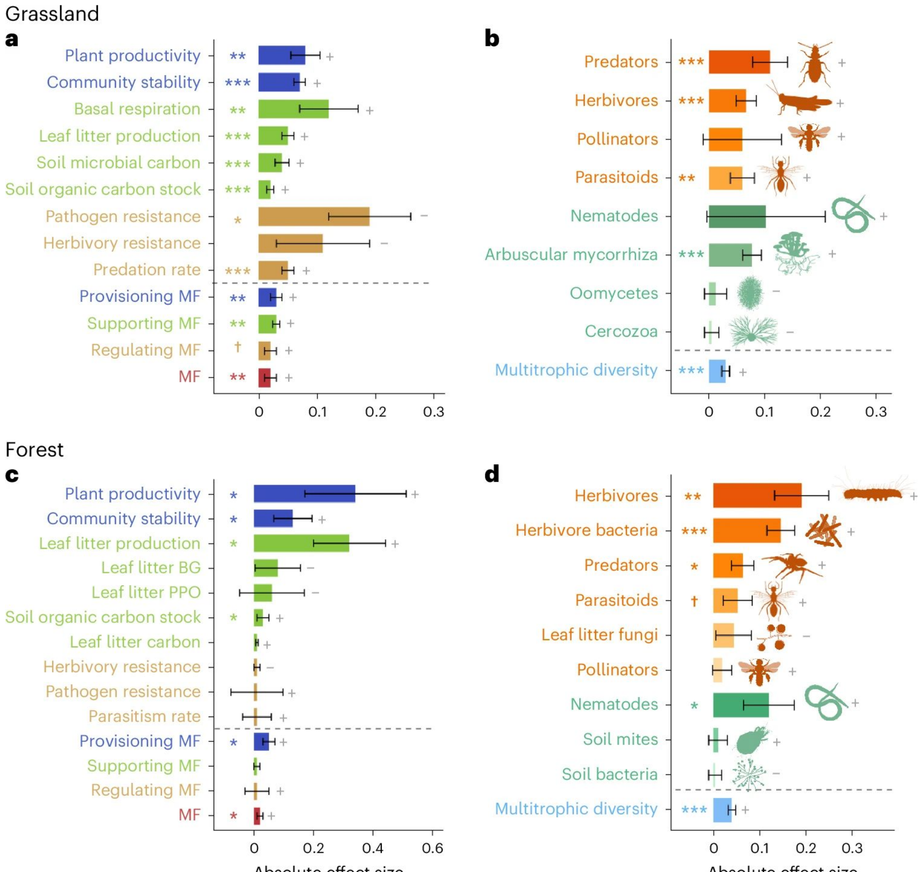
Check for updates

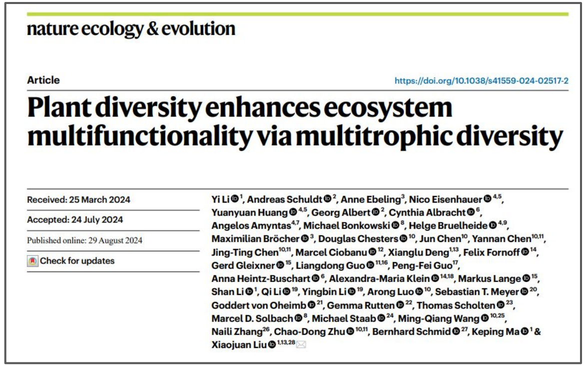
Yi Li[1], Andreas Schuldt[2], Anne Ebeling[3], Nico Eisenhauer[4,5], Yuanyuan Huang[4,5], Georg Albert[2], Cynthia Albracht[6], Angelos Amyntas[4,7], Michael Bonkowski[8], Helge Bruelheide[4,9], Maximilian Bröcher[3], Douglas Chesters[10], Jun Chen[10], Yannan Chen[10,11], Jing-Ting Chen[10,11], Marcel Ciobanu[12], Xianglu Deng[1,13], Felix Fornoff[14], Gerd Gleixner[15], Liangdong Guo[11,16], Peng-Fei Guo[17], Anna Heintz-Buschart[6], Alexandra-Maria Klein[14,18], Markus Lange[15], Shan Li[1], Qi Li[19], Yingbin Li[19], Arong Luo[10], Sebastian T. Meyer[20], Goddert von Oheimb[21], Gemma Rutten[22], Thomas Scholten[23], Marcel D. Solbach[8], Michael Staab[24], Ming-Qiang Wang[10,25], Naili Zhang[26], Chao-Dong Zhu[10,11], Bernhard Schmid[27], Keping Ma[1] & Xiaojuan Liu[1,13,28]





Ecosystem Multifunctionality - Plant diversity effects





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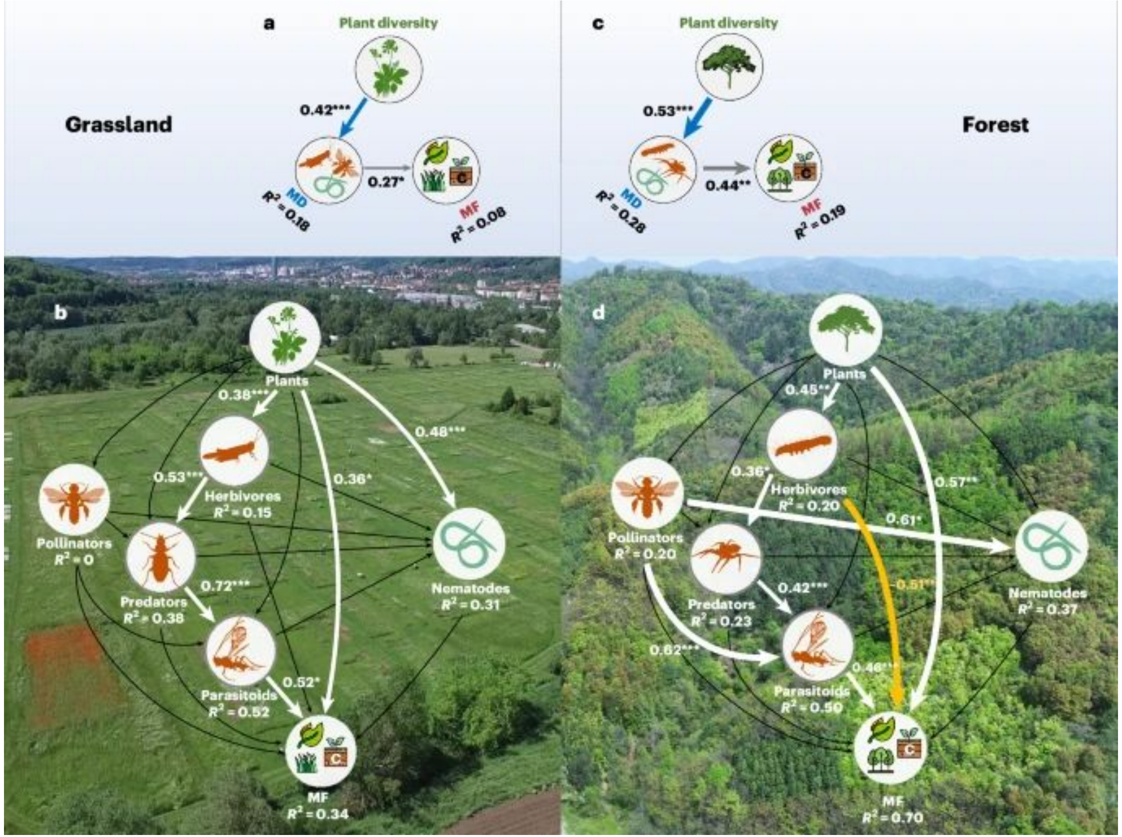
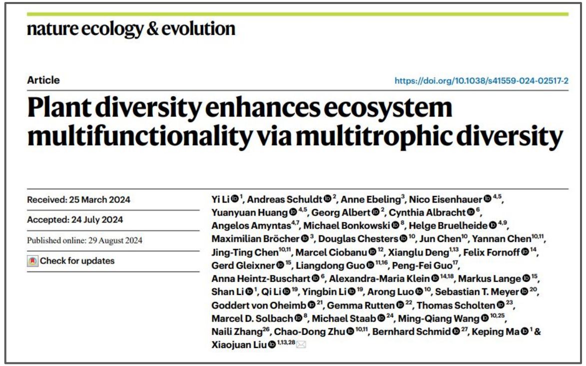
Ecosystem Multifunctionality - Trophic effects





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Next class: More Biodiversity and Ecosystem Functioning!